MAE 8001: Advanced Continuum Mechanics

Course Level:	Graduate (First-Year MS/PhD)
Credits:	3 (average 3 hours per week)
Meeting Time:	Every Tuesday, 7pm – 10 pm (approximately 3 × 50 minutes per week), 204 3rd teaching building
Instructor:	Jiarui Wang , 819 North Engineering Building, wangjr@sustech.edu.cn

Course Description

Topics include tensor analysis, kinematics of deformation, stress measures, balance laws, constitutive theory, and linear and finite elasticity. The course establishes the theoretical foundations for advanced studies in nonlinear solid mechanics and computational mechanics.

Course Objectives

1. Use tensor notation and tensor calculus in continuum formulations.
2. Describe finite deformation using appropriate kinematic measures.
3. Formulate stress tensors and interpret their physical meaning.
4. Derive conservation and balance laws in local form.
5. Apply principles of material frame indifference and isotropy.
6. Develop constitutive models for linear and hyperelastic materials.
7. Analyze simple boundary value problems in both small and finite deformation settings.

Weekly Schedule

Weeks 1–3: Vector and Tensor Analysis

- Vector algebra and inner/outer products
- Index notation and Einstein summation
- Kronecker delta and permutation tensor
- Principal values and invariants
- Tensor differentiation
- Gradient, divergence, material derivative, divergence theorem

Weeks 4–6: Kinematics of Finite Deformation

- Motion and deformation mapping
- Deformation gradient F
- Line, area, and volume change
- Right and left Cauchy–Green tensors
- Green–Lagrange and Almansi strain tensors
- Polar decomposition

- Deformation rate measures
- Small-strain approximation

Weeks 7–8: Stress Measures

- Traction vector
- Cauchy's theorem
- Cauchy stress tensor
- Symmetry of stress
- Kirchhoff, First and second Piola-Kirchhoff stresses
- Principal stresses and invariants

Weeks 9–10: Conservation and Balance Laws

- Conservation of mass
- Balance of linear momentum
- Balance of angular momentum
- Mechanical energy balance
- Stress power and internal work
- First and second law of thermodynamics

Weeks 11-12: Constitutive Theory

- Material frame indifference
- Dissipation inequality
- Isotropic material response

Week 13-14: Mechanics of solids

- Hyperelasticity
- Strain energy density
- Material moduli

Weeks 15: Nonlinear finite element implementation

- Total and updated Lagrangian weak forms
- Push-back and push forward
- Linearization
- Material and geometrical response in the linearized solids

Weeks 16: Final review

Grading Policy

- Homework: 35%
- Attendance: 10%
- Midterm Exam or Final Project: 15%
- Final Exam: 40%